

EX. NO: 8

IMPLEMENTATION OF DIFFIE–HELLMAN KEY EXCHANGE ALGORITHM

AIM

To write a C program to implement the Diffie–Hellman Key Exchange Algorithm.

ALGORITHM

Step 1: Start the program.

Step 2: Read the public prime number P and primitive root G.

Step 3: Read the private keys of Alice (a) and Bob (b).

Step 4: Compute Alice's public key:

$$A = G^a P$$

Step 5: Compute Bob's public key:

$$B = G^b P$$

Step 6: Exchange the public keys.

Step 7: Alice computes the shared secret key:

$$K_A = B^a P$$

Step 8: Bob computes the shared secret key:

$$K_B = A^b P$$

Step 9: Verify that both shared keys are equal.

Step 10: Display the public keys and the shared secret key.

Step 11: Stop the program.

PROGRAM:

```
#include <stdio.h>

// Function to calculate (base^exp) % mod
long long power(long long base, long long exp, long long mod)
{
    long long result = 1;

    while(exp > 0)
    {
        result = (result * base) % mod;
        exp--;
    }
}
```

```

    return result;
}

int main()
{
    long long P, G;
    long long a, b;
    long long A, B;
    long long keyA, keyB;
    printf("Enter Prime Number (P): ");
    scanf("%lld", &P);
    printf("Enter Primitive Root (G): ");
    scanf("%lld", &G);
    printf("Enter Alice's Private Key: ");
    scanf("%lld", &a);
    printf("Enter Bob's Private Key: ");
    scanf("%lld", &b);
    A = power(G, a, P);
    B = power(G, b, P);
    keyA = power(B, a, P);
    keyB = power(A, b, P);
    printf("\nAlice's Public Key : %lld", A);
    printf("\nBob's Public Key : %lld", B);
    printf("\nShared Secret Key (Alice): %lld", keyA);
    printf("\nShared Secret Key (Bob) : %lld\n", keyB);
    if(keyA == keyB)
        printf("\nKey Exchange Successful.\n");
    else
        printf("\nKey Exchange Failed.\n");
    return 0;
}

```

OUTPUT:

The screenshot displays the Programiz Online Compiler interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The page header includes the Programiz logo, a banner for Adelaide University, and a 'Programiz PRO' button. The main area is divided into a code editor on the left and an output window on the right.

Code Editor (main.c):

```
33 printf("Enter Bob's Private Key: ");
34 scanf("%lld", &b);
35
36 // Generate public keys
37 A = power(G, a, P);
38 B = power(G, b, P);
39
40 // Generate shared secret keys
41 keyA = power(B, a, P);
42 keyB = power(A, b, P);
43
44 printf("\nAlice's Public Key : %lld", A);
45 printf("\nBob's Public Key : %lld", B);
46
47 printf("\nShared Secret Key (Alice): %lld", keyA);
48 printf("\nShared Secret Key (Bob) : %lld\n", keyB);
49
50 if(keyA == keyB)
51     printf("\nKey Exchange Successful.\n");
52 else
53     printf("\nKey Exchange Failed.\n");
54
55 return 0;
56 }
```

Output:

```
Enter Prime Number (P): 23
Enter Primitive Root (G): 5
Enter Alice's Private Key: 6
Enter Bob's Private Key: 15

Alice's Public Key : 8
Bob's Public Key : 19
Shared Secret Key (Alice): 2
Shared Secret Key (Bob) : 2

Key Exchange Successful.

=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the system clock at 09:21 PM on 03-08-2026.